

0.1 SMOOTH NONCONVEX AND MATCHED-NOISE CONTROLS

The existing nonconvex campaign supplies a smooth SGD construction with a proved invariant law and separately measured positive tangent growth. Its batch loss is

$$L_B(x) = \frac{h_B x^2}{2} - \xi_B x - \frac{0.92}{2} [x^2 - \log(1 + x^2)], \quad \xi_B \mid h_B \sim \mathcal{N}(0, \sigma^2 h_B / B),$$

where h_B averages B independent equiprobable values 0.95 and 1.05. Every sampled loss is coercive, but the population Hessian attains -0.035 . The exact update is

$$x^+ = [1 - \eta(h_B - 0.92)]x - \frac{0.92\eta x}{1 + x^2} + \eta\xi_B.$$

At $\eta = 8$, the outer coefficient lies in $[-0.04, 0.76]$ and the nonlinear remainder is bounded. Global drift and Gaussian minorization therefore give a unique attracting invariant law with every polynomial moment finite, independently of the numerical tangent calculation. The reference at $x = 0$ has log gain 1.94427 and harmonic gain 6.97714. Three completed FP64 configurations, each with 16 chains and one million updates, give mean moving log growth $+0.18959$ for $B = 1$, $\sigma = 0.001$. These finite-horizon measurements suggest positive long-time growth but do not prove the smooth model’s infinite-time exponent; the positive stationary exponent of the piecewise control in the main text is established analytically.

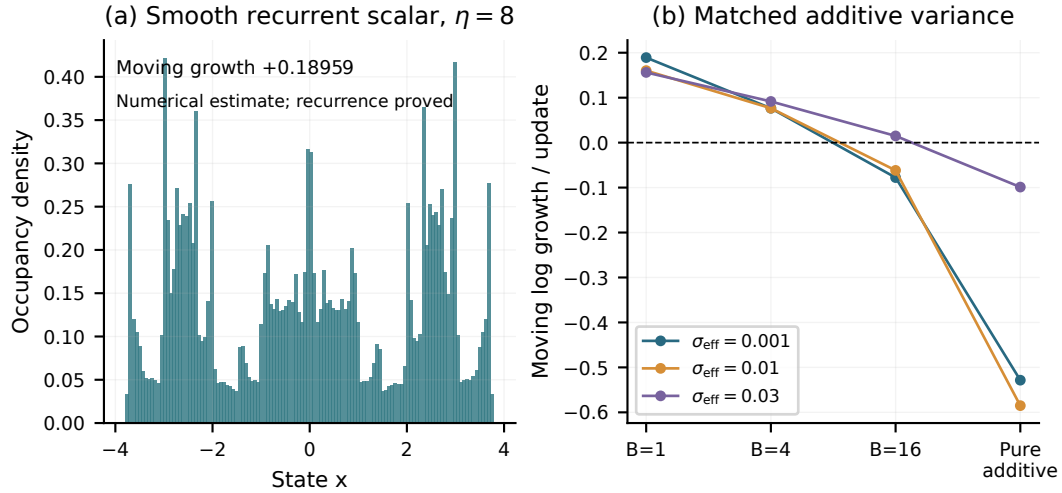


Figure 1: **Smooth recurrence and a controlled multiplicative-noise intervention.** (a) Pooled occupancy from all saved 20,000-update dense followups to the three million-update, 16-chain FP64 confirmations at $\eta = 8$, $B = 1$, $\sigma = 0.001$. The density is multimodal; no exactly-two-mode claim is made. The displayed moving growth is estimated from the primary long trajectories. (b) A separate intervention sets the per-example residual scale to $\sigma_{\text{eff}}\sqrt{B}$, so that $\text{Var}(\xi_B) = \sigma_{\text{eff}}^2$ stays fixed while $\text{Var}(h_B) = 0.0025/B$ changes. Points average completed million-update configurations; the pure-additive control sets $h_B = 1$. Thus batching can change the moving-growth sign even with matched unconditional additive variance.

The matched-variance results distinguish the two sources of randomness: at $\sigma_{\text{eff}} = 0.001$, moving growth changes from $+0.18917$ at $B = 1$ to -0.07775 at $B = 16$, whereas the pure-additive control gives -0.52845 . Conditional dependence between h_B and ξ_B remains as specified; matching variance does not make them independent. The finite-time means, correlated occupancy samples, and local density shape do not prove the sign of an asymptotic tangent exponent or a universal harmonic-to-bimodal transition.

0.2 CONSTRUCTED NEURAL RECURRENCE VERSUS ORDINARY-NETWORK CONTROLS

The constructed networks have two or four affine layers, hidden width four, and tanh or sigmoid activations, fitted to 64 fixed inputs sampled uniformly from $[-1, 1]^2$, with targets $\sin(3x_1) + 0.7x_1x_2 + 0.3\cos(4x_2)$. Effective weights and biases are bounded transforms of unconstrained latent parameters: $W = (2/\sqrt{\text{fan_in}}) \tanh \theta_W$, $b = \tanh \theta_b$. Their sampled objective is

$$\frac{1}{2B} \sum_{i \in I} (f_\theta(x_i) - y_i)^2 + \frac{\mu}{2} \|\theta\|^2 - \sigma_{\text{eff}} z^\top \theta, \quad z \sim \mathcal{N}(0, I).$$

Here the data gradient is globally bounded and the latent update has outer coefficient $1 - \eta\mu$; with $\mu = 0.05$ and $\eta \in \{0.5, 10\}$, recurrence with all moments follows from the global return argument. The full-rank parameter-linear forcing is an explicit design choice, distinct from regression label noise. Freezing the output bias restricts the state and forcing to the remaining active coordinates.

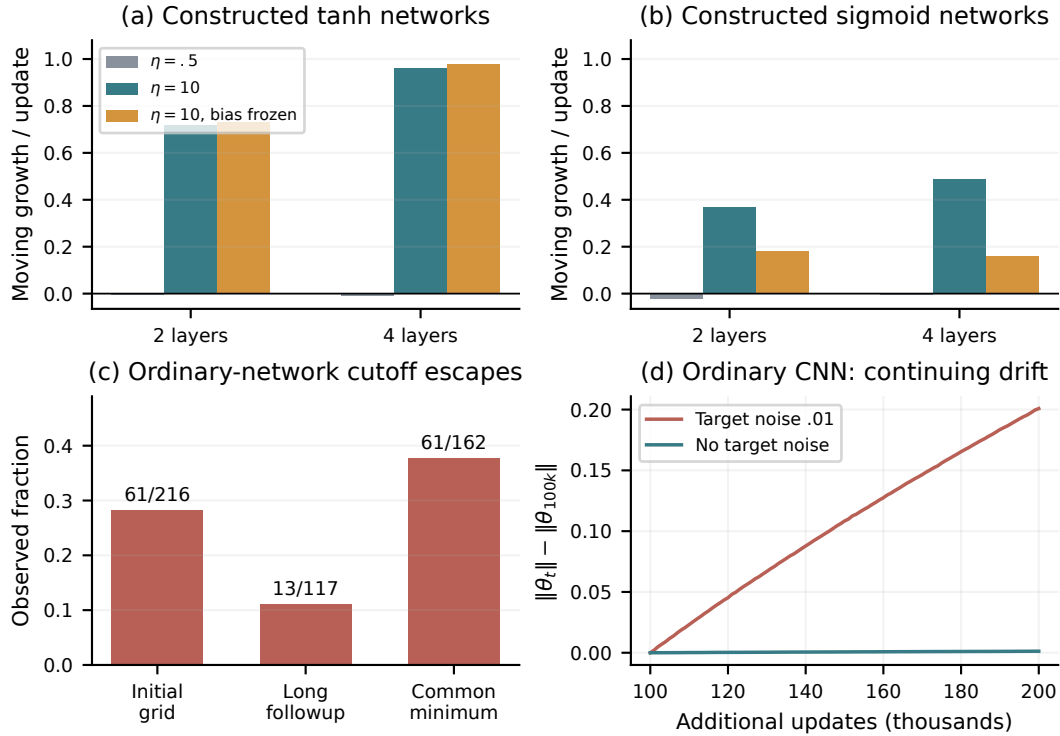


Figure 2: **Constructive positive controls do not imply ordinary-network equilibrium.** (a,b) At $B = 8$, $\sigma_{\text{eff}} = 0.003$, moving-growth means use the final 500,000 updates of completed million-update constructed-network confirmations, with 48 chains per setting. Large-step positive growth survives the frozen-output-bias ablation; four-layer tanh gives $+0.96022$ with trainable bias and $+0.97751$ with frozen bias. These are numerical growth estimates in models with separately proved recurrence. (c) The ordinary-network campaign retains all 495 completed cells, including 135 loss-cutoff escapes; cohorts have different designs and horizons, so these fractions are not a controlled comparison of escape probabilities. (d) A separate ordinary CIFAR-10 CNN at $\eta = 0.08$, $B = 128$ continues to drift over 200,000 additional updates. The displayed noisy and noiseless paths share their parent checkpoint and minibatch stream. A finite path’s norm drift does not measure ensemble-variance growth or prove mathematical escape.

In the matched CNN pair, the norm-mean shift between updates $[100000, 150000)$ and $[150000, 200000)$ is 0.10005 with target-noise standard deviation 0.01 , versus 0.000613 without added noise. Its tiny float32 moving-growth signs are unresolved and are not used to claim an invariant law. The broader ordinary controls include continuing parameter drift and dead-feature ReLU states whose remaining output bias follows a stable scalar recursion; these are not sustained positive-tangent neural equilibria. All figures use completed executions. The prospective next-study protocol contributes no results or counts.